

HiMCM 2022 Problem A — Summary Sheet

Team XXXX sources: `ccd_healthy_vs_stress_snapshot.csv`, `autograd_elasticity_ranking.csv`,
`ccd_tipping_point.csv`, `orchard_pollination_optimization.csv`

What we built. A five-state dual-feedback DFM colony ODE $y = [B, H, F, P, N]$ with C^∞ activations, unrolled in PyTorch for Autograd elasticities; a triple-stress CCD bifurcation in forager mortality μ_F ; and a 20-acre almond competition–yield–rent OR model. All figures in `paper_figures/` are generated from code; no hand-edited CSV cells.

Req 1 — Colony dynamics (`ccd_healthy_vs_stress_snapshot.csv`). Healthy baseline ($\mu_F = 0.10$): **summer peak adults** $H + F = 66,507$ (day 238); **overwinter adults** 38,622. Field triple stress at $\mu_F = 0.28$: summer peak 5,630, overwinter 3,163. Legacy Khoury-style series `colony_annual_dynamics.csv`: year peak adults 27,546 (DOY 32), winter end 2,074 (companion, not mixed into Torch peaks).

Req 2 — Autograd elasticity ranking (`autograd_elasticity_ranking.csv`). For $J_{\text{pop}} = H + F$ at day 365 ($J^* = 35,179$):

1. L_{max} (laying rate): $S = +0.925$
2. μ_F (forager mortality): $S = -0.311$
3. σ_F (social inhibition): $S = +0.294$

For $J_{\text{nectar}} = N$ at day 270: top is η_N with $S = +0.829$.

Req 3 / CCD — Stress tipping (`ccd_tipping_point.csv`). Under pesticide + Varroa + frost, **soft tip** $\mu_F^{*,\text{soft}} = 0.11$ (leave Safe Zone $N \geq 5000$ g); **hard collapse tip** $\mu_F^* = 0.305 \text{ day}^{-1}$ (N crosses 3000 g). Sweep: 6 Safe / 39 Compensatory / 20 Collapse points.

Req 3 — 20-acre almond stocking (`orchard_pollination_optimization.csv`). Golden band **1–2 hives/acre** $\Leftrightarrow K \in [20, 40]$ colonies on 20 acres. Band optimum $K^* = 40$: YieldRatio = **0.848** (84.8% of full set), net profit \$51,394, hive-health index 0.975. At $K = 35$ (1.75/acre): yield 0.831, net \$51,136. Overstocking to $K = 60$ keeps yield ≈ 0.855 but drops health to 0.735 and net to \$47,826.

Req 4 — Grower takeaway. Protect queen laying and keep effective forager death below the soft tip under combined stress; stock almonds in the 20–40 colony band; see `grower_advisory_letter.tex` for placement, feeding, and Varroa triggers.